Skyline (Urban / High-Rise / Air Reconnaissance & Vertical Battlefield)

Executive Summary

Skyline is a towering **high-rise urban battlefield** built for **air reconnaissance**, **vertical combat**, **rooftop warfare**, **and multi-level autonomous operations**. The terrain features skyscraper clusters, rooftop landing pads, elevated skybridges, interior atriums, ventilation corridors, commercial towers, and dense street-level grids. The map emphasizes Obsidian Protocol's autonomous vertical-navigation logic, air-ground coordination, sensor-relay networks, and logistics under extreme verticality. In AR (Nreal Light/Air), the map anchors at ~6–5 m viewing distance, covering a 10×10 m physical area representing ~1 km² of high-rise urban terrain. Key features include rooftop vantage points, skybridge chokepoints, multi-tier combat zones, and strategic control of airspace corridors.

Design Goals & Gameplay

Air Reconnaissance & Vertical Dominance
Skyline prioritizes aerial recon, rooftop overwatch, VTOL deployment, drone scouting, and vertical flanking. Airspace corridors between towers define movement and targeting.
Multi-Level Urban Combat
Combat unfolds across street level, mid-tower floors, rooftops, and skybridges. Units must navigate elevators, stairwells, maintenance shafts, and rooftop pads.
Autonomy & Vertical Pathfinding
Autonomous units choose between rooftop routes, interior corridors, skybridge traversal, or street-level infiltration. Units adapt to blocked floors and shifting airspace control.
Sensor & Information Warfare
Glass reflections, heat plumes, smoke, and vertical occlusion distort sensors. Relay towers on rooftops restore clarity. Destroying relays creates blind pockets across tower clusters.
Logistics & Resources
Commercial loading bays, rooftop pads, and maintenance depots serve as resupply nodes. Convoys must navigate narrow streets and avoid ambush-prone tower bases.
Deployment & Progress
A fixed deployment budget (~10,000 points) forces players to choose between air support, rooftop infantry, or heavy urban armor. Blue deploys from the southwest tower plaza; Red deploys from the northeast corporate spire.

Spatial Constraints (Nreal AR)

Device Capabilities
Nreal Light/Air: ~52° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.
Playable Area & Scaling
10×10 m physical area = 1 km² high-rise zone (1:100 scale). Designed for tabletop AR.
Tracking & Occlusion
Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System
1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.
Key Locations
Blue HQ (Tower Plaza Command) — (0,0). 150×150 m Rooftop pads, drone bays, armored staging area.
Red HQ (Corporate Spire Command) — (1000,1000). 150×150 m Skyscraper command node with radar mast and AA emplacements.
Central Skyline Cluster — (500,450). 300×300 m High-rise towers, skybridges, rooftop combat zones.
Commercial Loading Bay — (750,800). 100×150 m Cargo docks, trucks, interior flanking routes.
Atrium Tower Complex — (800,300). 200×200 m Multi-level interior combat, elevators, stairwells.
Skybridge Network — (0,450 → 400,900) Elevated traversal routes; critical vertical chokepoint.
StreetsLevel Grid — (300,0 → 450,1020) Dense urban streets; ideal for armor and autonomous squads.
Rooftop Relay A — (650,150). +280 m Lone-range sensor vantage point.
Rooftop Relay B — (150,650). +300 m Comms relay tower and drone hub.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Skybridge Network
Major vertical chokepoint; controlling it determines tower-to-tower movement.
Atrium Tower Complex
Interior vertical combat; elevators and stairwells create unpredictable engagements.
Skyline Cluster Rooftops
Long-range sightlines; ideal for air-ground coordination.
StreetsLevel Grid
Close-range combat; armor and autonomous infantry excel.
Sightlines
Rooftop Relay A provides long-range sensor coverage (~350 m). Glass reflections and smoke create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW tower plaza + rooftop pads.
Red Deployment: NE corporate spire + AA emplacements.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain
High-rises, skybridges, rooftops, atriums, street grids.
Vegetation
Planters, rooftop gardens, sparse trees.
Buildings & Props
Military: AA guns, radar dishes, watchtowers. Urban: Offices, apartments, commercial towers, skybridges. Industrial: Loading bays, generators, HVAC units. Logistics: Trucks, crates, drones, fuel drums.
Effects
Smoke, sparks, dust, heat shimmer, glass reflections.
Audio
Wind between towers, machinery, distant aircraft, radio chatter.

LOD & Budget

Asset Cate	Count	Tri/Item	Total Tris
High-Rise	3	200k	600k
Skybridge	4	80k	320k
Buildings	15	50k	750k
Roads/Pa	4	30k	120k
Vegetatio	120	2k	240k
Props	50	10k	500k
Total	—	—	**~2.53M**

Texture Budget
2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins
Unreal AR Template + Nreal SDK (NREAL). ARKit/ARCore enabled.
Rendering
Forward renderer, stationary lights, baked GI, aggressive culling.
Anchors
Single spatial anchor for entire map.
Interaction
Phone pointer for pan/zoom; HUD as screen-space widgets.
Multiplayer
Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones
1. Prototype Layout — Month 1 Block-out towers, skybridges, atriums. AR anchor test.
2. Core Systems — Month 2 NavMesh, vertical autonomy behaviors, air-ground coordination.
3. AR Integration — Month 3 Anchors, plane detection, basic interactivity.
4. Content Pa — Month 4–5 Final models, textures, lighting, audio.
5. Gameplay & Polish — Month 6–7 Objectives, balance, VFX (glass reflections, smoke).
6. Testing & QA — Month 8 Multiplayer stress test, sensor validation.

Testing Plan

Autonomy
Verify vertical navigation, rooftop logic, skybridge traversal.
Comms/Sensors
Test relay destruction, fog-of-war updates, LOS checks.
Performance
≥60 Hz AR render on target device.
QA Checklist
Navigation, vertical behavior, deployment budget enforcement, anchor drift, sensor accuracy.